

Notebook: week3_model_building.ipynb

Week 3 — Model Building

Project: AI-Powered Fake News Detection Using Text Classification

Phase: 3 of 4 (Days 15–21)

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This notebook covers:

- Loading the TF-IDF feature matrices and labels (`X\_train\_tfidf.joblib`, `y\_train.joblib`)
- Initializing the four classifier architectures with baseline hyperparameters:
 - **K-Nearest Neighbors** (KNN, non-parametric)
 - **Logistic Regression** (parametric)
 - **Random Forest** (ensemble)
 - **Multi-Layer Perceptron** (MLP, simple neural network)
- Training all four classifiers on the TF-IDF representation
- Measuring and comparing training times (in seconds)
- Serializing the fitted models to disk (`results/<name>.joblib`)
- Exporting training logs for the Phase 3 report

```
import sys
import os
from pathlib import Path
```

```
# Ensure project root is on path
PROJECT_ROOT = Path(".").resolve()
if PROJECT_ROOT.name == "notebooks":
    PROJECT_ROOT = PROJECT_ROOT.parent
```

```
if str(PROJECT_ROOT) not in sys.path:
    sys.path.insert(0, str(PROJECT_ROOT))
```

```
os.chdir(PROJECT_ROOT)
print(f"Working directory: {PROJECT_ROOT}")
```

[Output]:
Working directory: .

```
import joblib
import time
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

```
from src.utils import ensure_dirs
from src.models import get_models, train_all, save_models
```

```
ensure_dirs("data/processed", "results/figures", "results/metrics")
sns.set_theme(style="whitegrid", palette="muted")
plt.rcParams["figure.dpi"] = 120
```

1. Load Training Data

We load the stratified TF-IDF training split and the corresponding labels from `data/processed/`.

```
X_train_path = Path("data/processed/X_train_tfidf.joblib")
y_train_path = Path("data/processed/y_train.joblib")

if not X_train_path.exists() or not y_train_path.exists():
    raise FileNotFoundError(
        "TF-IDF training set or labels not found. Run week2_feature_engineering.ipynb first."
    )

X_train = joblib.load(X_train_path)
y_train = joblib.load(y_train_path)

print(f"Loaded TF-IDF training matrix of shape: {X_train.shape}")
print(f"Loaded labels array of shape: {y_train.shape}")

[Output]:
Loaded TF-IDF training matrix of shape: (31278, 5000)
Loaded labels array of shape: (31278,)
```

2. Model Training

We load the models with their fixed baseline hyperparameters:

- **KNN**: $K=5$, uniform weights.
- **Logistic Regression**: maximum iterations set to 1000 to prevent early stopping convergence warnings.
- **Random Forest**: 100 decision tree estimators.
- **MLP Classifier**: single hidden layer of 100 neurons, Adam solver, maximum 300 epochs.

Training times are recorded using high-precision performance counters.

```
models = get_models()
print("Initialised models:")
for name, model in models.items():
    print(f" {name}: {model}")

[Output]:
Initialised models:
KNN: KNeighborsClassifier()
LogReg: LogisticRegression(max_iter=1000, random_state=42)
RandomForest: RandomForestClassifier(random_state=42)
NeuralNet: MLPClassifier(max_iter=300, random_state=42)

print("Starting training for all four models...")
fitted_models = train_all(models, X_train, y_train)
print("Training process complete.")

[Output]:
Starting training for all four models...
Training KNN... done (0.0s)
Training LogReg... done (0.1s)
Training RandomForest... done (13.3s)
Training NeuralNet... done (30.3s)
Training process complete.
```

3. Training Time Comparison

We plot and analyze the time taken to train each classifier on the 31,278 document training set.

```
times_dict = {name: info[1] for name, info in fitted_models.items()}
times_df = pd.DataFrame(list(times_dict.items()), columns=["Model", "Training Time (s)"])
print("=== Training Time Summary ===")
```

```
print(times_df.to_string(index=False))

fig, ax = plt.subplots(figsize=(6, 4))
sns.barplot(data=times_df, x="Model", y="Training Time (s)", ax=ax, palette="viridis")
ax.set_title("Training Time Comparison")
ax.set_ylabel("Time (seconds)")
for i, v in enumerate(times_df["Training Time (s)"]):
    ax.text(i, v + (max(times_df["Training Time (s)"]) * 0.01), f"{v:.2f}s", ha="center", va="bottom")

plt.tight_layout()
plt.savefig("results/figures/training_times.png", dpi=150)
plt.show()
print("Saved: results/figures/training_times.png")
```

[Output]:

```
=== Training Time Summary ===
      Model  Training Time (s)
      KNN          0.026517
      LogReg        0.108838
RandomForest      13.313324
      NeuralNet     30.252218
```

[Output]:

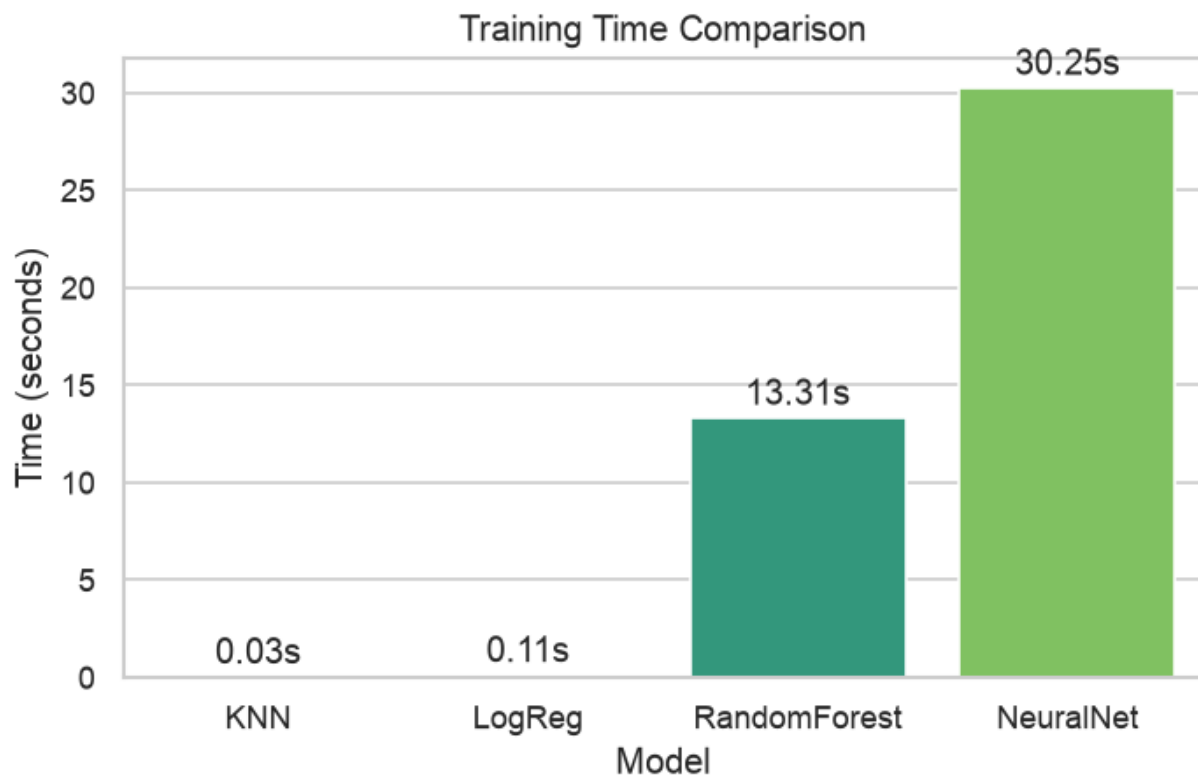
```
/var/folders/yv/w6571qrn2m74xpc6m96ljlzm0000gn/T/ipykernel_29930/294602422.py:7: FutureWarning:
```

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and set `order` to the order of the categories.

```
sns.barplot(data=times_df, x="Model", y="Training Time (s)", ax=ax, palette="viridis")
```

[Output]:

```
Saved: results/figures/training_times.png
```



4. Model Serialization

We serialize the fitted model weights so they can be loaded by the evaluation module in Phase 4.

```
save_models(fitted_models)
```

```
print("Saved all fitted model files (.joblib) to results/")
```

```
[Output]:
```

```
Saved: results/KNN.joblib
```

```
Saved: results/LogReg.joblib
```

```
Saved: results/RandomForest.joblib
```

```
Saved: results/NeuralNet.joblib
```

```
Saved all fitted model files (.joblib) to results/
```

5. Exporting Training Metrics

We save the precise training times to the metrics folder for use in the Phase 3 report.

```
joblib.dump(times_dict, "results/metrics/phase3_training_times.joblib")
```

```
print("Saved training times dictionary to results/metrics/phase3_training_times.joblib")
```

```
[Output]:
```

```
Saved training times dictionary to results/metrics/phase3_training_times.joblib
```

6. Hyperparameter Tuning

We perform a Grid Search Cross-Validation (3-fold) on Logistic Regression and Random Forest to find the best hyperparameters and compare them with the baseline defaults.

```
from src.models import tune_hyperparameters
```

```
tuned_info = tune_hyperparameters(X_train, y_train)
```

```
joblib.dump(tuned_info["LogReg"], "results/LogReg_tuned.joblib")
```

```
joblib.dump(tuned_info["RandomForest"], "results/RandomForest_tuned.joblib")
```

```
joblib.dump(tuned_info["LogReg_params"], "results/metrics/logreg_best_params.joblib")
```

```
joblib.dump(tuned_info["RandomForest_params"], "results/metrics/randomforest_best_params.joblib")
```

```
print("Tuning complete and tuned models saved.")
```